# ST. CECILIA'S COLLEGE-CEBU, INC.

*LASSO Supervised School*

*Ward II, Poblacion Minglanilla, Cebu*

# REAL ESTATE SYSTEM 2026

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## INTRODUCTION

This system is designed to help manage real estate transactions in a simple and organized way. Users can create accounts, log in, and access features for handling property listings and client information. The system stores data securely and makes it easy to update records when needed. It is user-friendly and helps make real estate tasks faster and more efficient.

## SIGNIFICANCE

The real estate system improves the way property transactions are handled by replacing manual processes with a digital system. It reduces errors, keeps records organized, and allows users to access information quickly. With this system, users can manage listings, track transactions, and update details without hassle. It also helps save time and effort, making work more efficient and reliable.

This system modernizes traditional real estate record-keeping by:

- Reducing Errors: Automated ID generation and data validation minimize manual entry mistakes.
- Enhancing Security: Role-based access (Admin vs. User) ensures sensitive management features are protected.
- Accountability: A built-in System Audit Log tracks every major action, including deletions and new registrations.
- Instant Access: Real-time updates allow users to see property availability (Available vs. Sold) immediately.

## BENEFACTORS

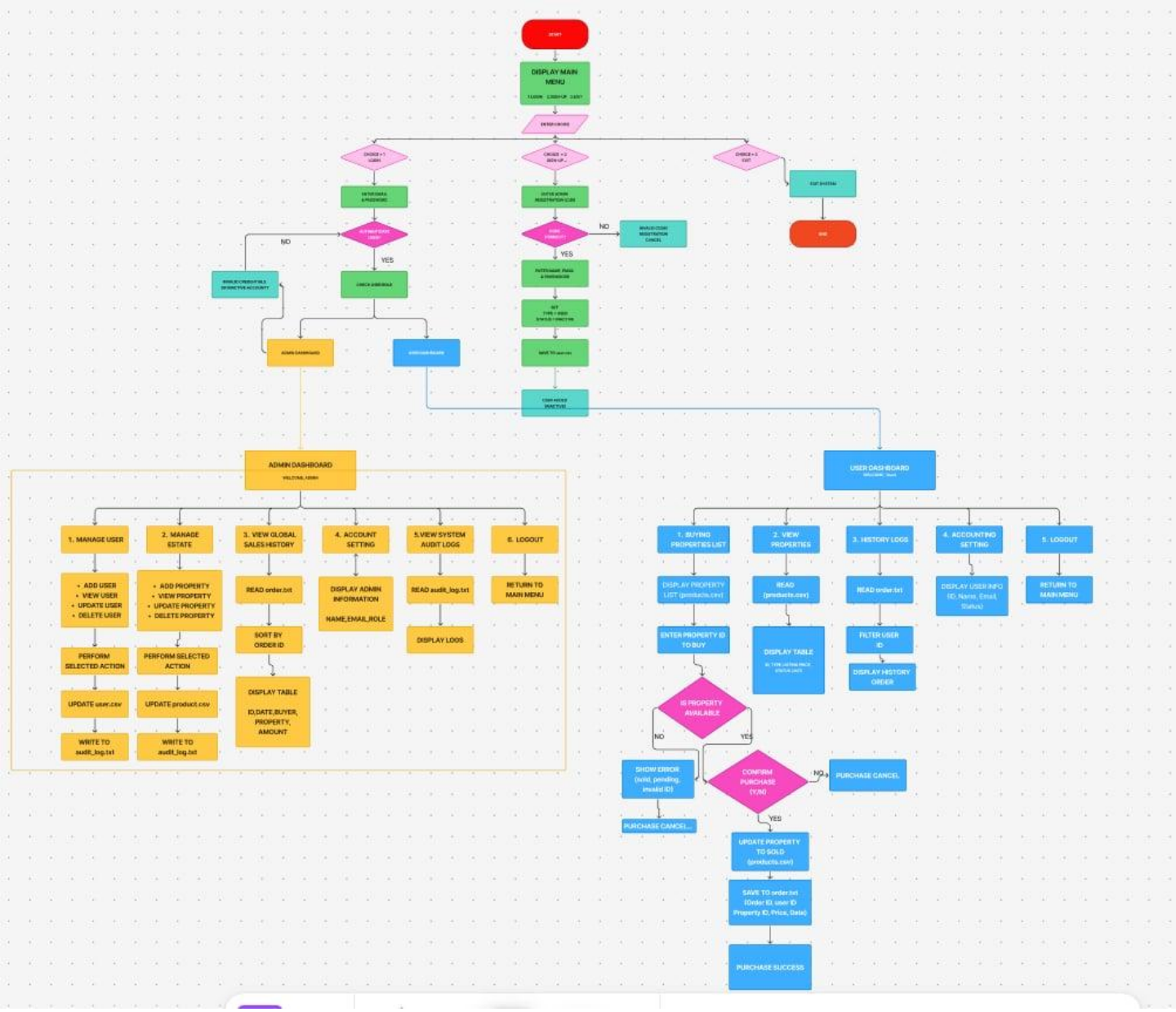
**1. Real Estate Agents / Admins**

- Easily manage property listings and client records.
- Reduce paperwork and avoid data loss.
- Quickly update property information and transactions.

**2. Users / Clients**

- Access property details anytime.
- Faster and more convenient transactions.
- Organized and accurate information.

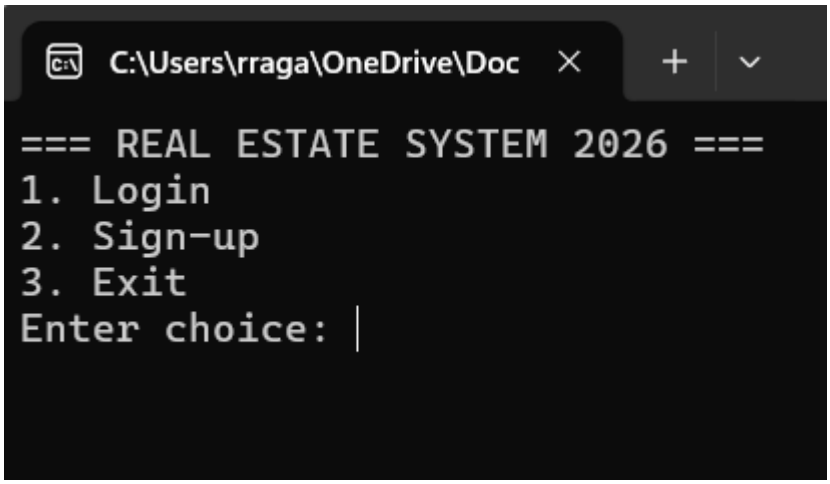
# FLOWCHART


START
DISPLAY MAIN MENU
NO
YES
NO
YES
ADMIN DASHBOARD
USER DASHBOARD
1. MANAGE USER
2. MANAGE ESTATE
3. VIEW GLOBAL SALES HISTORY
4. ACCOUNT SETTING
5.VIEW SYSTEM AUDIT LOGS
6. LOGOUT
• ADD USER
• VIEW USER
• UPDATE USER
• DELETE USER
• ADD PROPERTY
• VIEW PROPERTY
• UPDATE PROPERTY
• DELETE PROPERTY
READ order.txt
DISPLAY ADMIN INFORMATION
NAME,EMAIL,ROLE
READ audit_log.txt
RETURN TO MAIN MENU
PERFORM SELECTED ACTION
PERFORM SELECTED ACTION
SORT BY ORDER ID
DISPLAY LOGS
UPDATE user.csv
UPDATE product.csv
DISPLAY TABLE
ID,DATE,BUYER, PROPERTY, AMOUNT
WRITE TO audit_log.txt
WRITE TO audit_log.txt
1. BUYING PROPERTIES LIST
2. VIEW PROPERTIES
3. HISTORY LOGS
4. ACCOUNTING SETTING
5. LOGOUT
DISPLAY PROPERTY LIST (products.csv)
READ (products.csv)
READ order.txt
DISPLAY USER INFO (ID, Name, Email, Status)
RETURN TO MAIN MENU
ENTER PROPERTY ID TO BUY
DISPLAY TABLE
FILTER USER ID
DISPLAY HISTORY ORDER
IS PROPERTY AVAILABLE
NO
YES
SHOW ERROR (sold, pending, invalid ID)
CONFIRM PURCHASE (Y/N)
NO
PURCHASE CANCEL
PURCHASE CANCEL...
YES
UPDATE PROPERTY TO SOLD (products.csv)
SAVE TO order.txt (Order ID, user ID Property ID, Price, Date)
PURCHASE SUCCESS

# SYSTEM MANUAL

## MAIN DISPLAY

When you run the program, this is the 1st thing you will see




Select the option you want to operate

**Options**

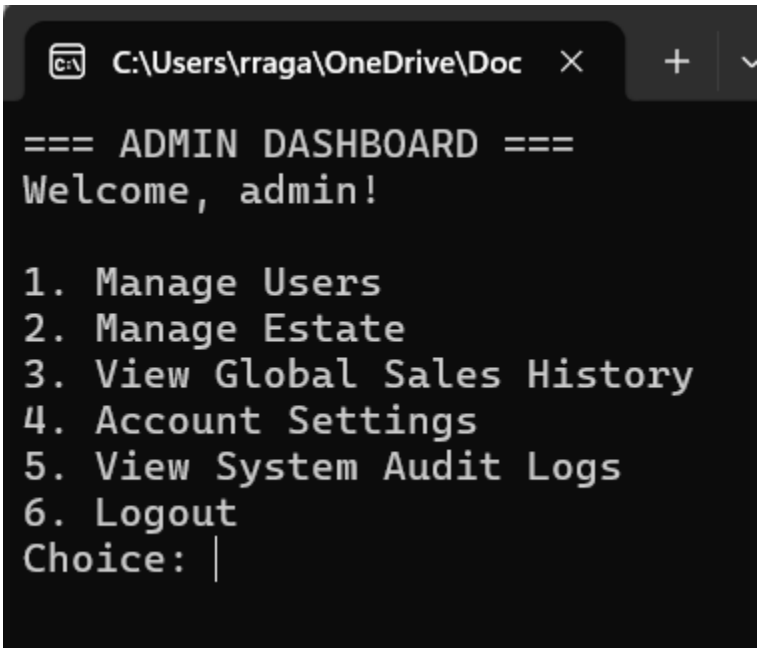
1. **Login - Existing users can enter their email and password to access their respective dashboards.**
2. **Sign-up: New users can register.** *Note:* Registering as an Admin requires a specific Registration Code for security**.**
3. **Exit: Securely closes the system and saves all pending CSV data.**

Note: If choice is 1 login it will detect if the account is Admin or User and will decide which dashboard to enter.

**If account equals to Admin it will choose Admin Dashboard**

### A. Admin Dashboard (Management Interface)

Upon a successful administrator login, the system grants access to the high-level management suite. This dashboard is the central hub for overseeing both personnel and property data.

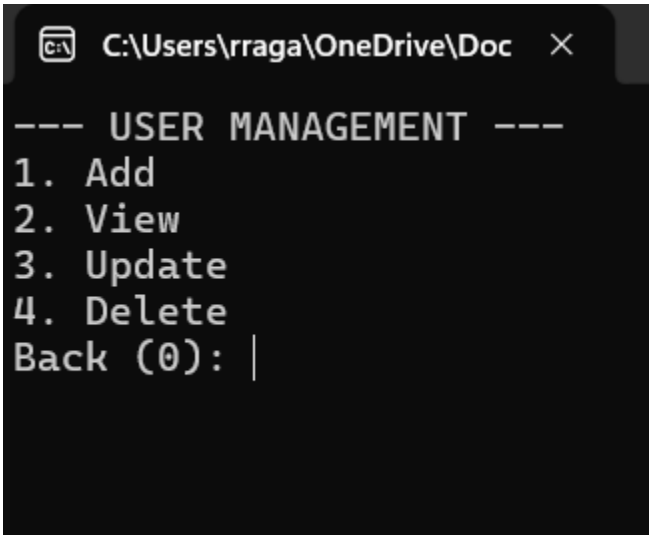



**Admin Functions**

1. Manage Users: Access the sub-menu to view all registered users, edit account details, or deactivate accounts.
2. Manage Estate: The primary interface for adding new property listings, updating prices, or modifying listing statuses (Sale/Rent).
3. View Global Sales History: A comprehensive report of every transaction made within the system across all users.
4. Account Settings: View and update your own administrator profile information.
5. View System Audit Logs: Review the security trail of the system, including property deletions and new user registrations with exact timestamps.
6. Logout: Securely terminates the admin session and returns to the Main Menu.

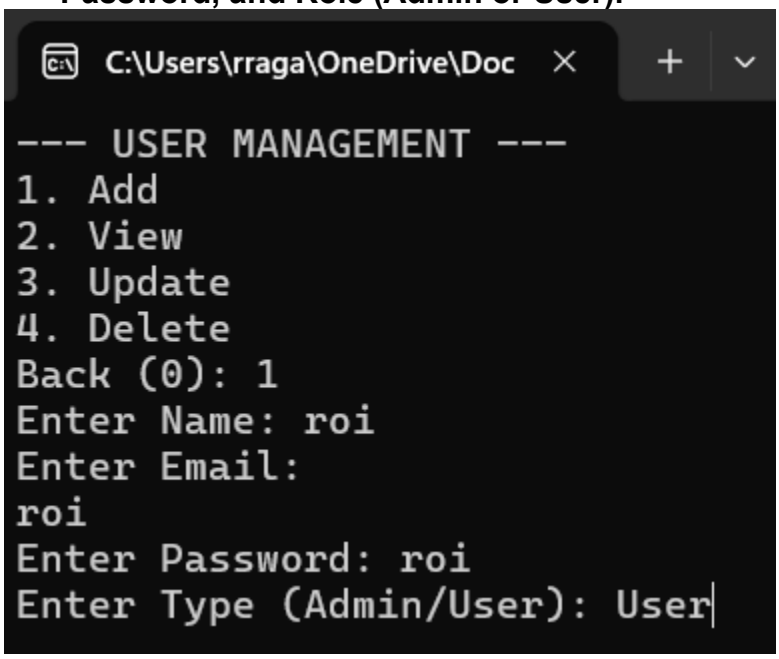
## B. Manage Users

This module allows administrators to perform CRUD (Create, Read, Update, Delete) operations on user accounts. Any changes made here are instantly updated in the users.csv file.




**OPERATIONS AVAILABE**

1. **Add: Manually register a new account. You will be prompted to enter a Name, Email, Password, and Role (Admin or User).**



2. **View: Displays a formatted table of all registered users, including their unique User ID and current account status.**

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=== USER MANAGEMENT LIST ===

| ID     | NAME              | EMAIL                  | TYPE      | STATUS      | REG DATE      |
|--------|-------------------|------------------------|-----------|-------------|---------------|
| 1      | admin             | admin                  | Admin     | Active      | 2026-04-04    |
| 2      | user              | user                   | User      | Active      | 2026-04-04    |
| 4      | ryle              | ryle                   | User      | Active      | 2026-04-04    |
| 5      | raganas           | raganas                | User      | Inactive    | 2026-04-10    |
| 6      | roi               | roi                    | User      | Inactive    | 2026-04-10    |
Press any key to continue . . .
```

3. **Update: Modify existing information for a specific user. You must provide the User ID of the account you wish to edit.**

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=== USER MANAGEMENT LIST ===

| ID     | NAME              | EMAIL                  | TYPE      | STATUS      | REG DATE      |
|--------|-------------------|------------------------|-----------|-------------|---------------|
| 1      | admin             | admin                  | Admin     | Active      | 2026-04-04    |
| 2      | user              | user                   | User      | Active      | 2026-04-04    |
| 4      | ryle              | ryle                   | User      | Active      | 2026-04-04    |
| 5      | raganas           | raganas                | User      | Inactive    | 2026-04-10    |
| 6      | roi               | roi                    | User      | Inactive    | 2026-04-10    |

Enter ID to Update: 6
New Name: roi
New Email: roi
New Type(Admin/User): User
New Status(Active/Inactive): Active

Update Successful!Press any key to continue . . .
```

4. **Delete: Remove a user profile from the system.** *(Warning:* This action is permanent and will be recorded in the System Audit Log.)

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=== USER MANAGEMENT LIST ===

| ID    | NAME            | EMAIL                | TYPE     | STATUS     | REG DATE     |
|-------|-----------------|----------------------|----------|------------|--------------|
| 1     | admin           | admin                | Admin    | Active     | 2026-04-04   |
| 2     | user            | user                 | User     | Active     | 2026-04-04   |
| 4     | ryle            | ryle                 | User     | Active     | 2026-04-04   |
| 5     | raganas         | raganas              | User     | Inactive   | 2026-04-10   |
| 6     | roi             | roi                  | User     | Active     | 2026-04-10   |

Enter ID to Delete: 6
Are you sure you want to delete User ID 6? (y/n): y

Property ID 6 deleted successfully.
User deleted successfully.
Press any key to continue . . .
```

5. **Back (0): Returns the user to the previous Admin Dashboard screen.**

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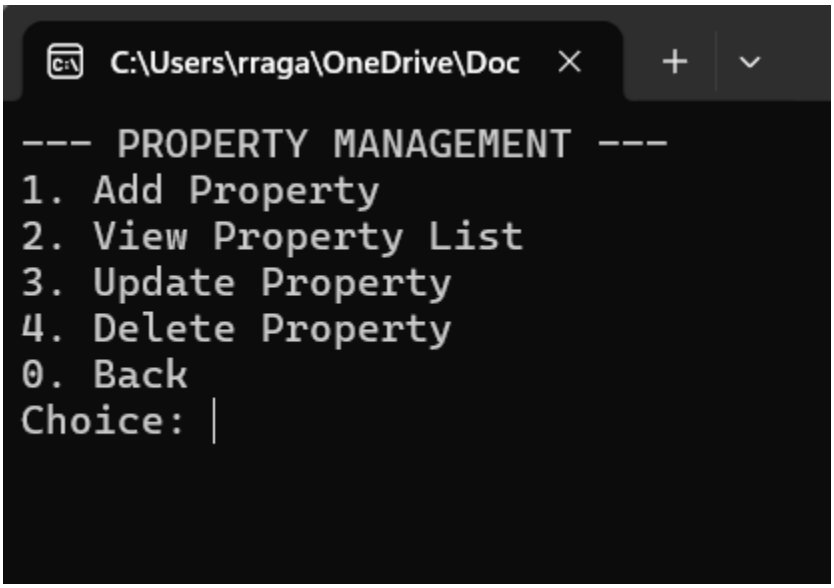
=== ADMIN DASHBOARD ===
Welcome, admin!

1. Manage Users
2. Manage Estate
3. View Global Sales History
4. Account Settings
5. View System Audit Logs
6. Logout
Choice:
```

## C. MANAGE ESTATE

### Property Management Sub-Menu

This module is dedicated to managing the lifecycle of property listings. Administrators use this menu to keep the marketplace up to date with new offerings or price adjustments.




**Available Operations:**

1. **Add Property: Create a new listing. You will be asked to input the Property Type (e.g., house, condo, lot), Listing Status (Rent or Sale), and the Price.**

2. **View Property List: Generates a full table of all properties currently in the system, displaying their unique Property ID, current Status (Available/Sold), and the Date Added.**

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--- PROPERTY MANAGEMENT ---
1. Add Property
2. View Property List
3. Update Property
4. Delete Property
0. Back
Choice: 2

| ID     | PROPERTY TYPE          | LISTING STATUS   | PRICE       | STATUS      | DATE ADDED    |
|--------|------------------------|------------------|-------------|-------------|---------------|
| 4      | fish                   | Rent             | 1.00        | Sold        | 2026-04-04    |
| 6      | house                  | Rent             | 5000.00     | Available   | 2026-04-10    |
| 7      | BUILDING               | Sale             | 100000.00   | Available   | 2026-04-10    |
Press any key to continue . . .
```

3. **Update Property: Allows for the modification of price or listing type for an existing entry. You must know the specific Property ID to proceed.**

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--- PROPERTY MANAGEMENT ---
1. Add Property
2. View Property List
3. Update Property
4. Delete Property
0. Back
Choice: 3

| ID     | PROPERTY TYPE          | LISTING STATUS   | PRICE       | STATUS      | DATE ADDED    |
|--------|------------------------|------------------|-------------|-------------|---------------|
| 4      | fish                   | Rent             | 1.00        | Sold        | 2026-04-04    |
| 6      | house                  | Rent             | 5000.00     | Available   | 2026-04-10    |
| 7      | BUILDING               | Sale             | 100000.00   | Available   | 2026-04-10    |

Enter Property ID to Update: 7
New Property Type: BUILDING
New Listing (1-Sale, 2-Rent, 3-Lease): 3
New Price: 45000
New Avail (1-Avail, 2-Sold): 1

Property ID 7 updated successfully!
Press any key to continue . . .
```

4. **Delete Property: Removes a listing from the database**. *Note:* This action will be logged in the System Audit Log to maintain a record of inventory changes.
   - 

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--- PROPERTY MANAGEMENT ---
1. Add Property
2. View Property List
3. Update Property
4. Delete Property
0. Back
Choice: 4

| ID     | PROPERTY TYPE          | LISTING STATUS   | PRICE       | STATUS      | DATE ADDED    |
|--------|------------------------|------------------|-------------|-------------|---------------|
| 4      | fish                   | Rent             | 1.00        | Sold        | 2026-04-04    |
| 6      | house                  | Rent             | 5000.00     | Available   | 2026-04-10    |
| 7      | BUILDING               | Lease            | 45000.00    | Available   | 2026-04-10    |

Enter Property ID to Delete: 7
Are you sure? (y/n): Y

Property ID 7 deleted successfully.
Press any key to continue . . .
```

5. **Back: Returns the administrator to the main Admin Dashboard.**

```
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=== ADMIN DASHBOARD ===
Welcome, admin!

1. Manage Users
2. Manage Estate
3. View Global Sales History
4. Account Settings
5. View System Audit Logs
6. Logout
Choice:
```

## D. View Global Sales History

The View Global Sales History provides a comprehensive log of every transaction processed by the system. This report is essential for financial auditing and tracking property turnover across all users.

**Information Displayed:**

- **ORD ID:** The unique tracking number automatically generated for every successful purchase.
- **DATE:** The timestamp (YYYY-MM-DD) recording exactly when the transaction occurred.
- **BUYER:** The username of the account that initiated the purchase.
- **PROPERTY:** The type of property sold (e.g., house, fish).
- **AMOUNT:** The total monetary value of the sale.
- **Total Sales:** A summary count at the bottom of the list showing the total number of transactions recorded in the database.

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=== GLOBAL SALES HISTORY ===

| ORD ID    | DATE         | BUYER           | PROPERTY    | AMOUNT      |
|----------|--------------|-----------------|------------|------------|
| 260      | 2026-04-04   | raganas         | house      | $100.00    |
| 459      | 2026-04-10   | user            | fish       | $1.00      |
| 656      | 2026-04-04   | user            | fish       | $250.00    |
| 955      | 2026-04-07   | user            | house      | $50.00     |

Total Sales: 4
Press any key to continue . . .
```

## E. My Account Details

Displays User ID, Name, Email, Role, and Status (Active/Inactive).

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=== MY ACCOUNT DETAILS ===
Name: admin
Email: admin
Role: Admin
Press any key to continue . . .
```

### F. System Audit Log

The **System Audit Log** serves as a transparent security trail for the entire application. It automatically records significant administrative actions and system events, ensuring accountability for every change made to the database.

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=== SYSTEM AUDIT LOG ===

[2026-04-04 15:58:49] [ryle      ] USER_ADDED      | New User Created: ID 4, Name: ryle
[2026-04-04 19:25:52] [gen       ] USER_ADDED      | New User Created: ID 5, Name: gen
[2026-04-05 15:59:25] [admin     ] PROP_DELETED    | Property Deleted: ID 5
[2026-04-07 11:31:46] [admin     ] PROP_DELETED    | Property Deleted: ID 5
[2026-04-07 11:32:16] [admin     ] PROP_DELETED    | Property Deleted: ID 3
[2026-04-10 15:13:05] [ryle      ] USER_ADDED      | New User Created: ID 5, Name: ryle
[2026-04-10 15:14:16] [admin     ] PROP_DELETED    | Property Deleted: ID 5
[2026-04-10 15:15:10] [admin     ] PROP_DELETED    | Property Deleted: ID 3
[2026-04-10 15:20:21] [test      ] USER_ADDED      | New User Created: ID 5, Name: test
[2026-04-10 15:20:31] [admin     ] PROP_DELETED    | Property Deleted: ID 5
[2026-04-10 15:34:18] [admin     ] PROP_DELETED    | Property Deleted: ID 7
[2026-04-10 15:36:30] [raganas   ] USER_ADDED      | New User Created: ID 5, Name: raganas
[2026-04-10 16:27:53] [roi       ] USER_ADDED      | New User Created: ID 6, Name: roi
[2026-04-10 16:31:14] [admin     ] PROP_DELETED    | Property Deleted: ID 6
[2026-04-10 16:40:14] [admin     ] PROP_DELETED    | Property Deleted: ID 7

Press any key to continue . . .
```

**Understanding Log Entries**

Each entry in the log is formatted to provide four critical pieces of information:

- **Timestamp:** The exact date and time (YYYY-MM-DD HH:MM:SS) the action occurred.
- **Name:** The username of the account that performed the action
- **Event Type:** The category of the action performed:
  - USER_ADDED: A new account was successfully registered in the system.
  - PROP_DELETED: A property listing was permanently removed from the inventory.
- **Action Details:** A brief description providing specific context, such as the **ID** and **Name** of the user created, or the unique **ID** of a deleted property.

### G. LOGOUT

**Return to Main Menu which is the REAL ESTATE SYSTEM 2026 Dashboard.**

# If login Account is User enter User Dashboard

## G.USER DASHBOARD

Once a standard user logs in, they are directed to the User Dashboard. This simplified interface is designed for quick property browsing and efficient purchasing.

```
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=== USER DASHBOARD ===
Welcome, user!

1. Buying Properties List
2. View Properties
3. History Logs
4. Account Settings
5. Logout
Choice:
```

**Dashboard Options:**

1. **Buying Properties List: Access the marketplace to select and purchase available properties.**

```
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=== PROPERTY LISTINGS ===

| ID     | PROPERTY TYPE         | LISTING STATUS  | PRICE       | STATUS      | DATE ADDED    |
|--------|-----------------------|-----------------|-------------|-------------|---------------|
| 4      | fish                  | Rent            | 1.00        | Sold        | 2026-04-04    |
| 6      | house                 | Rent            | 5000.00     | Available   | 2026-04-10    |

Enter Property ID to Buy: 6

--- PURCHASE CONFIRMATION ---
Buying: house
Price:  $5000.00
Are you sure? (y/n): y

 Order #749 saved.
 => SUCCESS! The property is now SOLD.
Press any key to continue . . .
```

2. **View Properties: View all properties in the system (including those already sold) for reference.**

```
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--- PROPERTIES ---

| ID     | PROPERTY TYPE          | LISTING STATUS   | PRICE       | STATUS      | DATE ADDED    |
|--------|------------------------|------------------|-------------|-------------|---------------|
| 4      | fish                   | Rent             | 1.00        | Sold        | 2026-04-04    |
| 6      | house                  | Rent             | 5000.00     | Sold        | 2026-04-10    |

Press any key to continue . . .
```

3. **History Logs: Review a personal record of all your past successful purchases.**

```
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=== YOUR PURCHASE HISTORY ===

Date: 2026-04-04 | Order #656   | Product ID: 4   | Total: $250.00
Date: 2026-04-07 | Order #955   | Product ID: 2   | Total: $50.00
Date: 2026-04-10 | Order #459   | Product ID: 4   | Total: $1.00
Date: 2026-04-10 | Order #749   | Product ID: 6   | Total: $5000.00

Press any key to continue . . .
```

4. **Account Settings: View your profile details and account status.**

```
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=== MY ACCOUNT DETAILS ===
User ID:   2
Name:      user
Email:     user
Status:    Active
-------------------------
Press any key to continue . . .
```

5. **Logout: Securely exit the user session.**

**H.SIGN-UP**

**This is the registration portal. Select this to create a new profile.**

- ***Note:*** **You will be asked to choose between a Client or Admin role.**
- **Registering as an Admin requires a System Security Code to prevent unauthorized access to management tools**

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=== REAL ESTATE SYSTEM 2026 ===
1. Login
2. Sign-up
3. Exit
Enter choice: 2

--- Sign-up ---
Enter Admin Registration Code: 1234
Enter Name: test
Enter Email: test
Enter Password: test

USER ADDED! ID: 6 | Date: 2026-04-10Press any key to continue . . .
```

**I. Exit**

**This option terminates the program execution. Always use this to ensure the system closes all data streams and saves the .csv files correctly.**

```
C:\Users\rraga\OneDrive\Doc

=== REAL ESTATE SYSTEM 2026 ===
1. Login
2. Sign-up
3. Exit
Enter choice: 3

Exiting System 2026. Goodbye!

--------------------------------
Process exited after 4631 seconds with return value 0
Press any key to continue . . .
```